

BADIS MOALLA

Software Test & Validation Engineer | Problem Manager | Automotive ECU | BMW Group

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PROFESSIONAL SUMMARY

Automotive Software Test & Validation Engineer with 3+ years of experience at KPIT Engineering within the BMW Group ecosystem, specialising in ECU validation, automotive security domains (certificate installation, token verification, ECU modes, DTC/diagnostics), and data-driven process improvement. Technical lead for the security test domain, working across AUTOSAR environments with Test Guide (TG). Proven ability to bridge development and testing teams, track team deviations via Jira dashboards, and automate reporting workflows using Python — cutting manual analysis time by 40% and incident resolution time by 25%. Trilingual: Arabic (native), English & French (C1). Available for immediate relocation to the GCC.

KEY ACHIEVEMENTS

-40% Manual analysis time via Python automation	-25% Incident resolution time via data-driven insights	100k+ ECU & defect records analyzed for root causes
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TECHNICAL SKILLS

ECU & Testing	AUTOSAR · Test Guide (TG) · ECU certificate installation · Token verification · ECU modes · DTC & diagnostics · Manual & automated testing
Automation	Python (Pandas, NumPy, Matplotlib) · Scripted defect & deviation analysis · Workflow automation · Jira API
Data & BI	Power BI (DAX, Power Query) · Jira dashboards & analytics · SQL · Excel (Power Query, Pivot Tables)
Process & QA	Root-cause analysis (RCA) · KPI definition & tracking · Continuous improvement · Agile / Scrum
Tools	Jira · Test Guide (TG) · Git · VS Code · MobaXterm · ELK Stack
Languages	Arabic (Native) · English (C1 – Fluent) · French (C1 – Fluent) · German (Basic)

PROFESSIONAL EXPERIENCE

Problem Manager – Data & Process Improvement

KPIT Engineering (BMW Group Automotive) · Sfax, Tunisia

May 2025 – Present

- ▶ Central liaison between development and testing teams: hear both sides of recurring issues, facilitate joint diagnosis, and drive resolution to closure.
- ▶ Monitor team deviations in real time using custom Jira dashboards built from scratch — providing management with instant visibility into backlog health and SLA compliance.
- ▶ Write Python scripts to automate deviation tracking and data checks that previously required hours of manual work, reducing analysis time by 40%.
- ▶ Analyze incident and quality datasets (100k+ records) to identify root causes, then implement fixes through Jira workflow changes, automated alerts, or process redesign.
- ▶ Manage end-to-end client reclamations: inform the team, investigate failure origin, and put preventive measures in place (scripts, dashboards, updated processes) to avoid recurrence.
- ▶ Facilitate continuous improvement meetings and track corrective actions through to full stakeholder sign-off.

Software Test & Validation Engineer – ECU Security Domain Lead

KPIT Engineering (BMW Group Automotive) · Sfax, Tunisia

Feb 2023 – May 2025

- ▶ Served as technical lead for the BMW ECU security test domain: planned, coordinated, and executed validation of certificate installation, token verification, ECU access modes, and secure boot sequences.
- ▶ Designed and executed manual and automated test cases for embedded automotive systems within AUTOSAR architecture using Test Guide (TG).
- ▶ Performed DTC (Diagnostic Trouble Code) and diagnostics job validation to ensure correct fault storage, readout, and clearing behaviour per BMW specifications.
- ▶ Analyzed large-scale test and defect datasets (100k+ records) to surface failure patterns and prepare quality risk reports for management and BMW stakeholders.
- ▶ Built Power BI dashboards tracking defect rate, test coverage, and release quality KPIs — used in weekly project reviews.
- ▶ Managed test data configurations and traceability matrices ensuring full repeatability across release cycles.

Data Analyst / BI Developer – Intern

AT Worthy (Remote) · Virginia, USA

Mar 2022 – Oct 2022

- ▶ Designed interactive Power BI and Tableau dashboards integrating multi-source business data for executive reporting.
- ▶ Built ETL pipelines using SSIS for data extraction, transformation, and loading across heterogeneous data sources.
- ▶ Standardised KPI definitions and datasets, improving reporting consistency and scalability.

BI Developer – Intern

Clinisys · Sfax, Tunisia

Jun 2020 – Sep 2020

- ▶ Built real-time monitoring dashboards using Power BI and Python for operational performance tracking.
- ▶ Designed SQL scripts for ad-hoc reporting and automated report generation.

NOTABLE PROJECTS

Automotive QA Deviation Tracker *Python · SQL · Power BI · Jira*

- ▶ Analyzed 6 months of QA & defect data (50k+ records) to identify top root causes; built interactive Power BI dashboard replacing weekly manual Excel reports.
- ▶ Reduced incident resolution time by 25% through data-driven recommendations adopted by the team lead.

Reporting Automation Pipeline *Python · Jira API · Excel*

- ▶ Built Python scripts to automatically extract Jira deviation data, clean it, and generate structured reports — eliminating 3–4 hours of manual work per week.
- ▶ Adopted by the team as the standard process for deviation tracking and client reporting.

EDUCATION

Diploma in Industrial Computer Engineering (Bac+5 – Engineering Degree)

ENET'Com – National School of Electronics and Telecommunications · Sfax, Tunisia

Sep 2019 – Dec 2022

CERTIFICATIONS

- ▶ Microsoft Power BI Data Analyst (PL-300) – In progress (2026)
- ▶ Google Data Analytics Professional Certificate
- ▶ Python Developer Certificate
- ▶ Scrum Fundamentals Certificate
- ▶ Linux Fundamentals Certificate